

Strict Reservation Visual Objectives

Baseline: symmetric behavior, $\lambda_1=\lambda_2=25$, $S=32$, $H=14$, seeds 5101-5105

Purpose

This memo is exploratory and visual, not a policy recommendation. It compares strict Class 1 reservation to pooled FCFS for different values of Q . It uses booked-slot utilization.

Objective Functions

In this memo, utilization means booked slots divided by available slots. The first function is weighted utilization:

$$U(Q) = w_1 \rho_1(Q) + w_2 \rho_2(Q)$$

The second function is weighted offered waiting time:

$$T(Q) = [w_1 \sum \tau_{\text{offered},1}(Q) + w_2 \sum \tau_{\text{offered},2}(Q)] / [w_1 \text{offered}_1(Q) + w_2 \text{offered}_2(Q)]$$

Current weights: $w_1=1.5$, $w_2=1$.

Served-rate flag: yes when the lower class served rate is below 50%.

Current Parameters

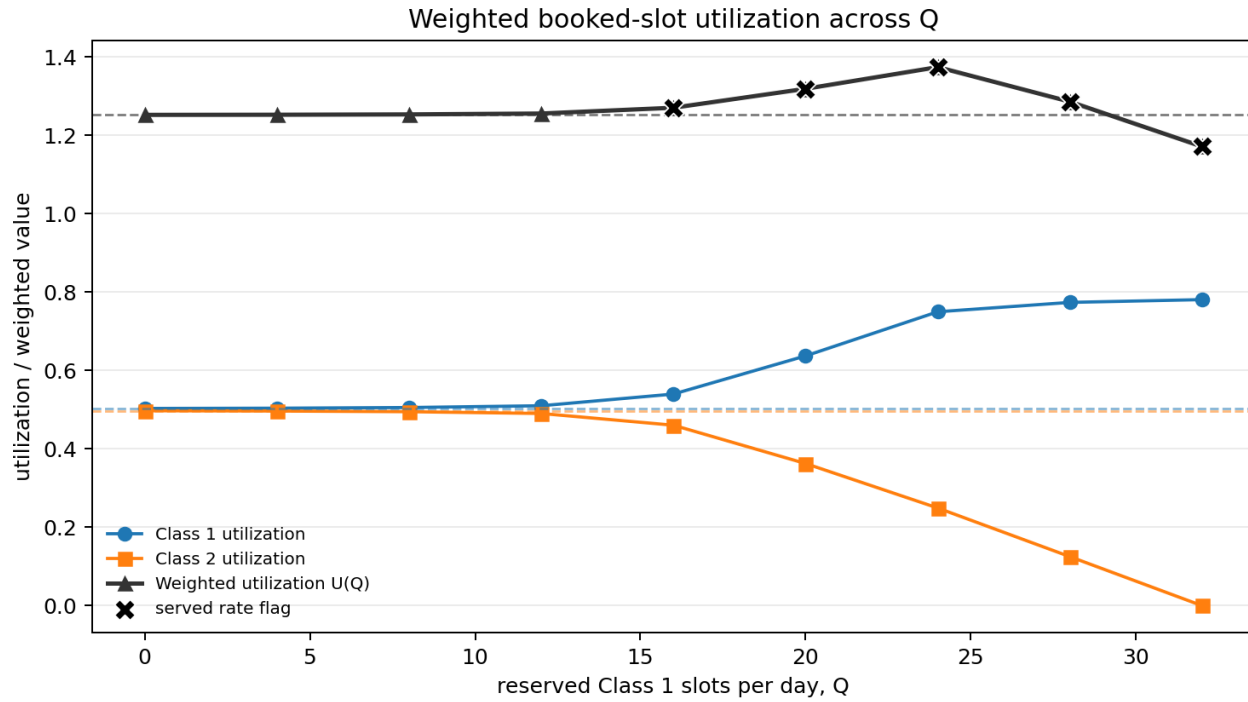
Scenario: symmetric baseline. Demand: $\lambda_1=\lambda_2=25$. Capacity: $S=32$ slots per day. Horizon: $H=14$ days. Seeds: 5101-5105. Q grid: 0, 4, 8, 12, 16, 20, 24, 28, 32.

How To Read The Plots

Dashed horizontal lines show the pooled FCFS reference. On the served-rate plot, black X markers are placed on the class line that is below 50%.

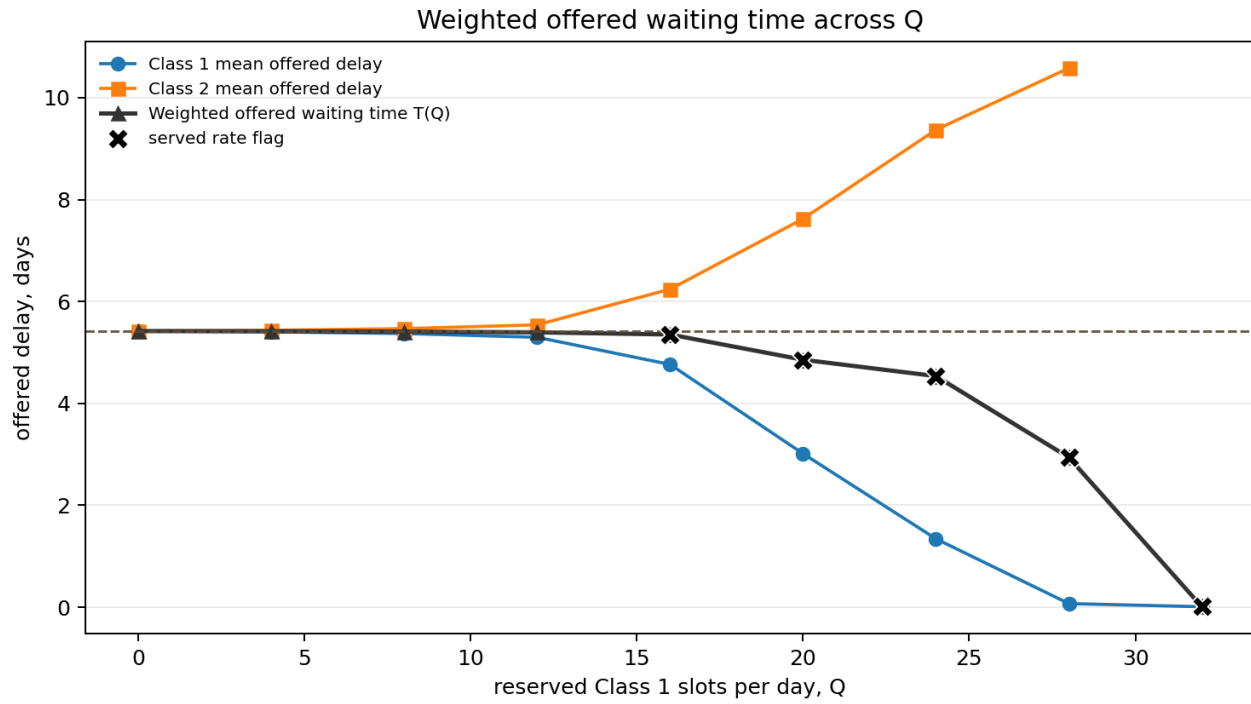
Weighted offered waiting time must be read together with served rates. A low offered waiting time is not necessarily good if it happens because one class stops receiving offers.

Figure 1. Weighted booked-slot utilization



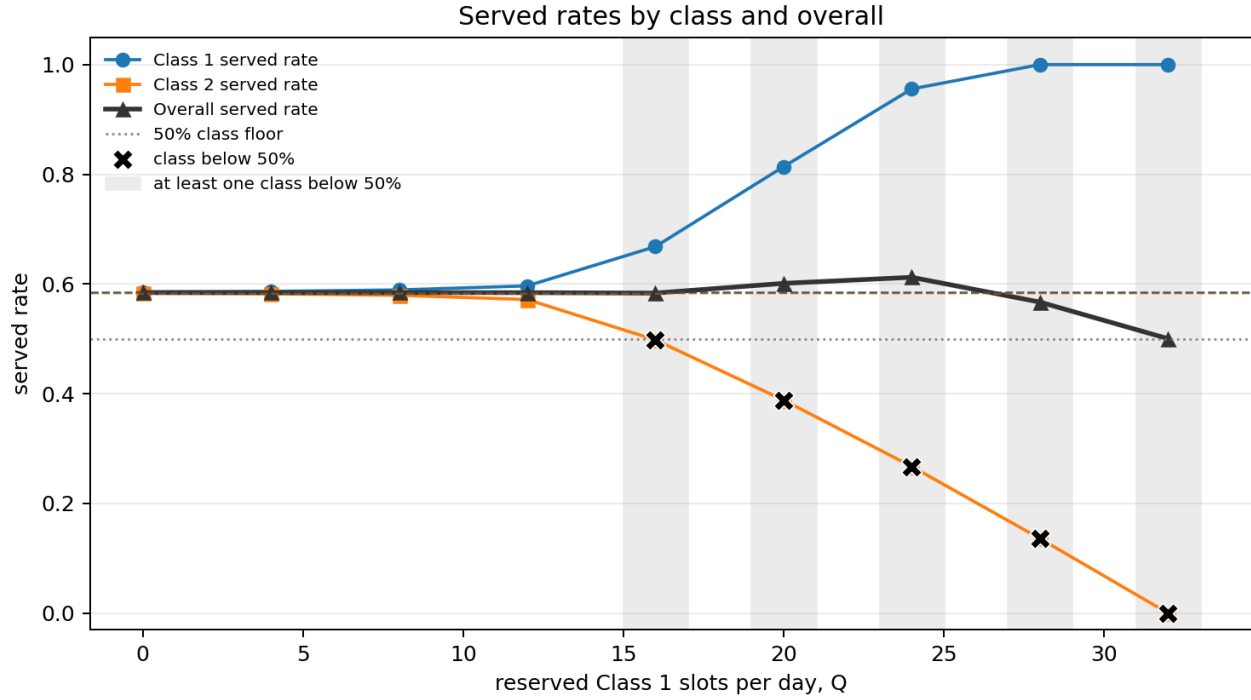
Class lines show raw utilization by class. The dark line shows $U(Q)$. Dashed lines show the pooled FCFS reference.

Figure 2. Weighted offered waiting time



Class lines show mean offered delay by class. The dark line shows $T(Q)$. Missing class delay values mean that class had no offered patients in that slice.

Figure 3. Served rates



This plot shows why the waiting-time objective must be read with served rates. The overall served rate stays near or above 50%, but Class 2 service falls below 50% as Q increases.

Served-Rate Flag Table

Q	Class 1 served rate	Class 2 served rate	minimum class served rate	served-rate flag
0	0.586	0.583	0.582	no
4	0.587	0.582	0.582	no
8	0.589	0.579	0.579	no
12	0.597	0.572	0.572	no
16	0.668	0.498	0.498	yes
20	0.814	0.388	0.388	yes
24	0.956	0.267	0.267	yes
28	1.000	0.136	0.136	yes
32	1.000	0.000	0.000	yes

Small Data Snapshot

Q	weighted utilization $U(Q)$	weighted offered waiting time $T(Q)$	Class 1 offered delay	Class 2 offered
0	1.253	5.419	5.420	5.418
4	1.253	5.416	5.404	5.434
8	1.254	5.409	5.371	5.468
12	1.256	5.395	5.298	5.542
16	1.271	5.352	4.764	6.241
20	1.319	4.857	3.014	7.623
24	1.374	4.535	1.341	9.371
28	1.286	2.938	0.070	10.581
32	1.172	0.009	0.009	

Next Step

Repeat the same visual analysis for equal arrival rates $\lambda_1=\lambda_2$ in 20, 25, 35, and 50.